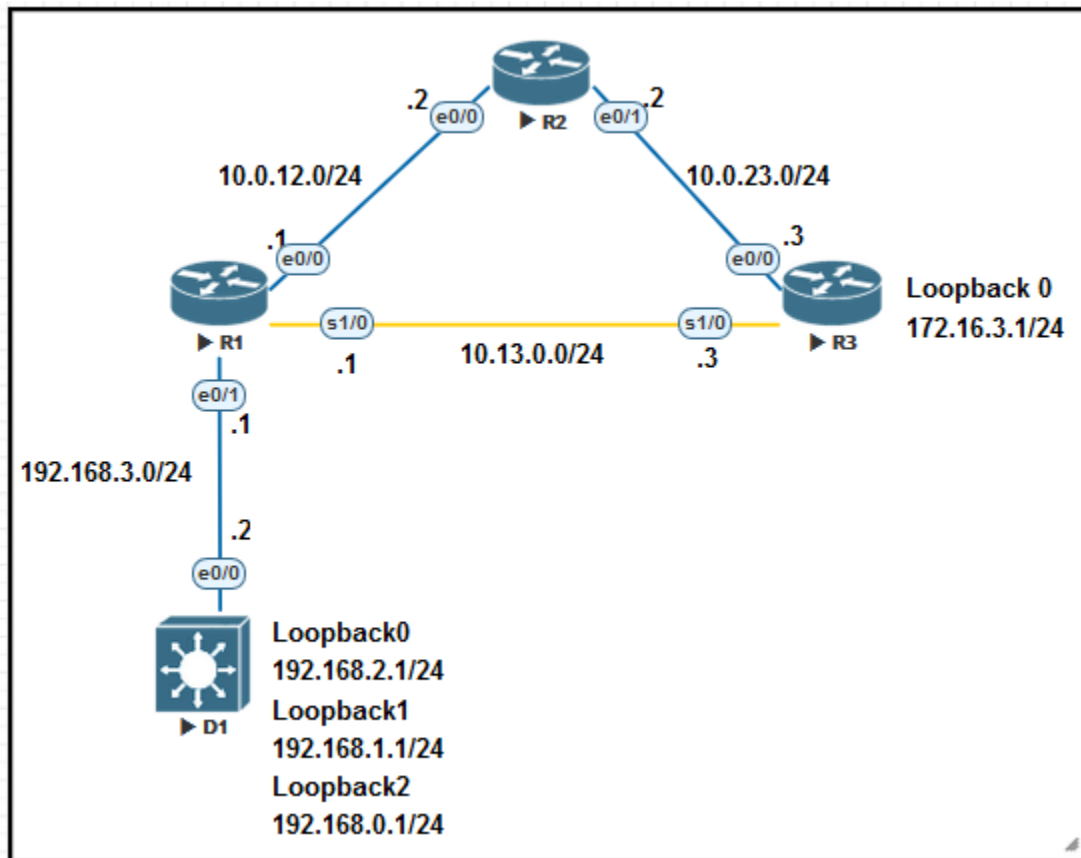


Lab - Implement Advanced EIGRP for IPv4 Features Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask
R1	E0/0	10.12.0.1	255.255.255.0
	E0/1	192.168.3.1	255.255.255.0
	S1/0	10.13.0.1	255.255.255.0
R2	E0/0	10.12.0.2	255.255.255.0
	E0/1	10.23.0.2	255.255.255.0
R3	E0/0	10.23.0.3	255.255.255.0
	S1/0	10.13.0.3	255.255.255.0
	Loopback0	172.16.3.1	255.255.255.0
D1	E0/0	192.168.3.2	255.255.255.0
	Loopback0	192.168.1.1	255.255.255.0

Device	Interface	IP Address	Subnet Mask
	Loopback1	192.168.1.1	255.255.255.0
	Loopback2	192.168.0.1	255.255.255.0

Objectives

Part 1: Build the Network and Configure Basic Device Settings

Part 2: Implement EIGRP for IPv4

Part 3: Implement Advanced Features

Background / Scenario

Customizing the operation of EIGRP can yield many benefits, most notably speeding convergence and stabilizing network operations during outages. In this lab you will explore some advanced techniques that can be used to customize and improve EIGRP performance on an enterprise network.

Part 1: Build the Network and Configure Basic Device Settings

Step 1: Configure basic settings for each device.

- Console into each device, enter global configuration mode, and apply the basic settings. The startup configurations for each device are provided below.

Router R1

```
hostname R1
no ip domain lookup
banner motd # R1, Implement Advanced EIGRP for IPv4 Features #
line con 0
  exec-timeout 0 0
  logging synchronous
  exit
line vty 0 4
  privilege level 15
  exec-timeout 0 0
  password cisco123
  login
  exit
!
interface e0/0
  ip address 10.12.0.1 255.255.255.0
  no shutdown
  exit
interface S1/0
  ip address 10.13.0.1 255.255.255.0
  no shutdown
  exit
interface e0/1
```

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```
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
end
```

Router R2

```
hostname R2
no ip domain lookup
banner motd # R2, Implement Advanced EIGRP for IPv4 Features #
line con 0
  exec-timeout 0 0
  logging synchronous
  exit
line vty 0 4
  privilege level 15
  exec-timeout 0 0
  password cisco123
  login
  exit
!
interface e0/0
  ip address 10.12.0.2 255.255.255.0
  no shutdown
  exit
interface e0/1
  ip address 10.23.0.2 255.255.255.0
  no shutdown
  exit
end
```

Router R3

```
hostname R3
no ip domain lookup
banner motd # R3, Implement Advanced EIGRP for IPv4 Features #
line con 0
  exec-timeout 0 0
  logging synchronous
  exit
line vty 0 4
  privilege level 15
  exec-timeout 0 0
  password cisco123
  login
  transport input telnet
  exit
interface e0/0
```

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```
ip address 10.23.0.3 255.255.255.0
no shutdown
exit
interface S1/0
ip address 10.13.0.3 255.255.255.0
no shutdown
exit
interface loopback 0
ip address 172.16.3.1 255.255.255.0
no shutdown
exit
end
```

Switch D1

```
hostname D1
no ip domain lookup
ip routing
banner motd # D1, Implement Advanced EIGRP for IPv4 Features #
line con 0
exec-timeout 0 0
logging synchronous
exit
line vty 0 4
privilege level 15
exec-timeout 0 0
password cisco123
login
exit
interface E0/0
no switchport
ip address 192.168.3.2 255.255.255.0
no shutdown
exit
interface loopback 0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
interface loopback 1
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface loopback 2
ip address 192.168.0.1 255.255.255.0
no shutdown
exit
end
```

- b. Set the clock on all devices to UTC time.
- c. Save the running configuration to startup-config on all devices.

Part 2: Implement EIGRP for IPv4

In this part, you will configure classic EIGRP for IPv4 and verify that all routing tables are converged.

Step 1: Configure classic EIGRP for IPv4.

Configure classic EIGRP for IPv4 on all devices. Use Autonomous System number 98, and advertise only the connected interfaces on each device.

Step 2: Verify EIGRP for IPv4 routing.

Verify that each device has a complete routing table for all the networks shown in the topology and Addressing Table.

Part 3: Implement Advanced Features

In this part of the lab you will customize several different settings within EIGRP and see the impact of those changes on the network.

Step 1: Modify timers.

EIGRP uses standard **hello-interval** and **hold-time** timers based on the speed of the interface. If the interface speed is a T1 or less, hellos are sent every 60 seconds and the hold-time is set to 180 seconds. If the interface speed is greater than a T1, hellos are sent every 5 seconds and the **hold-time** is set to 15 seconds. These default times might not be appropriate for some network scenarios.

- a. To see what the timers are set to, issue the **show ip eigrp interfaces detail** command.

```
R1# show ip eigrp interfaces detail
EIGRP-IPv4 Interfaces for AS(98)

              Xmit Queue   PeerQ           Mean    Pacing Time    Multicast
Pending
Interface      Peers  Un/Reliable  Un/Reliable  SRTT    Un/Reliable  Flow
Timer  Routes
Et0/0          1      0/0         0/0         1      0/050        0
Hello-interval is 5, Hold-time is 15
Split-horizon is enabled
Next xmit serial <none>
Packetized sent/expedited: 5/1
Hello's sent/expedited: 29/2
Un/reliable mcasts: 0/5  Un/reliable ucasts: 7/3
Mcast exceptions: 0  CR packets: 0  ACKs suppressed: 0
Retransmissions sent: 2  Out-of-sequence rcvd: 0
Topology-ids on interface - 0
Authentication mode is not set
Topologies advertised on this interface:  base
Topologies not advertised on this interface:

S1/0           1      0/0         0/0         1      0/1250        0
Hello-interval is 5, Hold-time is 15
Split-horizon is enabled
Next xmit serial <none>
```

Lab - Implement Advanced EIGRP for IPv4 Features

```
Packetized sent/expedited: 4/0
Hello's sent/expedited: 32/2
Un/reliable mcasts: 0/0  Un/reliable ucasts: 6/5
Mcast exceptions: 0  CR packets: 0  ACKs suppressed: 0
Retransmissions sent: 0  Out-of-sequence rcvd: 0
Topology-ids on interface - 0
Authentication mode is not set
Topologies advertised on this interface:  base
Topologies not advertised on this interface:
```

```
Et0/1          1          0/0          0/0          4          0/050          0
```

```
Hello-interval is 5, Hold-time is 15
Split-horizon is enabled
Next xmit serial <none>
Packetized sent/expedited: 2/1
Hello's sent/expedited: 31/2
Un/reliable mcasts: 0/2  Un/reliable ucasts: 4/3
Mcast exceptions: 0  CR packets: 0  ACKs suppressed: 0
Retransmissions sent: 2  Out-of-sequence rcvd: 0
Topology-ids on interface - 0
Authentication mode is not set
Topologies advertised on this interface:  base
Topologies not advertised on this interface:
```

- b. For this lab, the timers on R1 interface E0/0 and S1/0 need to be adjusted to send hellos every 10 seconds and establish a hold time of 30 seconds. EIGRP is unique in that each interface can have a customized **hello-interval** and **hold-time**. The times are not needed to match between the ends of a link. Change the timers using the **ip hello-interval eigrp ASN seconds** and the **ip hold-time eigrp ASN seconds** interface configuration commands.

```
R1# config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)# interface e0/0
R1(config-if)# ip hello-interval eigrp 98 10
R1(config-if)# ip hold-time eigrp 98 30
R1(config-if)# exit
R1(config)# interface S1/0
R1(config-if)# ip hello-interval eigrp 98 10
R1(config-if)# ip hold-time eigrp 98 30
R1(config-if)# exit
R1(config)# end
```

- c. To verify that the changes were made, check the output of the **show ip eigrp interfaces detail** command.

```
R1# show ip eigrp interfaces detail
EIGRP-IPv4 Interfaces for AS(98)

                                Xmit Queue   PeerQ           Mean    Pacing Time    Multicast
Pending
Interface      Peers  Un/Reliable    Un/Reliable    SRTT    Un/Reliable    Flow
Timer    Routes
Et0/0          1      0/0           0/0           1      0/050          0
```

Lab - Implement Advanced EIGRP for IPv4 Features

```
Hello-interval is 10, Hold-time is 30
```

```
Split-horizon is enabled
Next xmit serial <none>
Packetized sent/expedited: 5/1
Hello's sent/expedited: 84/2
Un/reliable mcasts: 0/5  Un/reliable ucasts: 7/3
Mcast exceptions: 0  CR packets: 0  ACKs suppressed: 0
Retransmissions sent: 2  Out-of-sequence rcvd: 0
Topology-ids on interface - 0
Authentication mode is not set
Topologies advertised on this interface:  base
Topologies not advertised on this interface:
```

```
S1/0          1          0/0          0/0          1          0/1250          0
```

```
Hello-interval is 10, Hold-time is 30
```

```
Split-horizon is enabled
Next xmit serial <none>
Packetized sent/expedited: 4/0
Hello's sent/expedited: 87/2
Un/reliable mcasts: 0/0  Un/reliable ucasts: 6/5
Mcast exceptions: 0  CR packets: 0  ACKs suppressed: 0
Retransmissions sent: 0  Out-of-sequence rcvd: 0
Topology-ids on interface - 0
Authentication mode is not set
Topologies advertised on this interface:  base
Topologies not advertised on this interface:
```

```
E10/1         1          0/0          0/0          4          0/050          0
```

```
Hello-interval is 5, Hold-time is 15
```

```
Split-horizon is enabled
Next xmit serial <none>
Packetized sent/expedited: 2/1
Hello's sent/expedited: 92/2
Un/reliable mcasts: 0/2  Un/reliable ucasts: 4/3
Mcast exceptions: 0  CR packets: 0  ACKs suppressed: 0
Retransmissions sent: 2  Out-of-sequence rcvd: 0
Topology-ids on interface - 0
Authentication mode is not set
Topologies advertised on this interface:  base
Topologies not advertised on this interface:
```

Step 2: Create summarized routes in EIGRP.

Large routing tables take more memory and require more CPU time to process. Reducing the size of the routing table is advantageous in all network scenarios. EIGRP supports summarization of routes at any point in the network. There is no boundary router limitation like the limitations imposed in OSPF. However, in order for the summary route to be valid, EIGRP requires that some component of the summary route be in the routing table for the router doing the summarization.

- a. Issue the **show ip route eigrp | begin Gateway** command on R3 and note the group of networks from the 192.168 range of addresses. R1 is advertising this contiguous block of networks individually, instead of sending a summary.

```
R3# show ip route eigrp | begin Gateway
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D      10.12.0.0/24 [90/307200] via 10.23.0.2, 00:04:40, Ethernet0/0
D      192.168.0.0/24 [90/460800] via 10.23.0.2, 00:04:10, Ethernet0/0
D      192.168.1.0/24 [90/460800] via 10.23.0.2, 00:04:13, Ethernet0/0
D      192.168.2.0/24 [90/460800] via 10.23.0.2, 00:04:15, Ethernet0/0
D      192.168.3.0/24 [90/332800] via 10.23.0.2, 00:04:40, Ethernet0/0
close configuration window
```

- b. On R1, configure a summary of the networks between R1 and D1, as well as the networks on D1 on the interfaces connecting to R3 and R2.

```
R1# conf t
R1(config)# interface e0/0
R1(config-if)# ip summary-address eigrp 98 192.168.0.0 255.255.252.0
*Mar  6 00:07:33.742: %DUAL-5-NBRCHANGE: EIGRP-IPv4 98: Neighbor 10.12.0.2
(Ethernet0/0) is resync: summary configured
R1(config-if)# exit
R1(config)# interface S1/0
R1(config-if)# ip summary-address eigrp 98 192.168.0.0 255.255.252.0
*Mar  6 00:07:35.220: %DUAL-5-NBRCHANGE: EIGRP-IPv4 98: Neighbor 10.13.0.3
(Serial0/1/0) is resync: summary configured
R1(config-if)# end
```

- c. Now examine the routing table on R3 again using the **show ip route eigrp | begin Gateway** command. In the output, you now see a single route taking the place of what had been four distinct routes.

```
R3# show ip route eigrp | begin Gateway
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D      10.12.0.0/24 [90/3072] via 10.23.0.2, 00:07:59, Ethernet0/0
D      192.168.0.0/22 [90/332800] via 10.23.0.2, 00:00:25, Ethernet0/0
```

Step 3: Control EIGRP query propagation with EIGRP stub routers.

EIGRP uses query messages to find a path to networks in the autonomous system. The query messages always require an acknowledgement. But a router will only send a response if it has a potential route that satisfies the query. If it does not have a route, it sends its own queries to its neighbors. This process can lead to long delays in reconvergence after an outage.

Query scoping refers to using various techniques to control how far across a network queries have to be sent. Summarization is one way of controlling query propagation. Another way to control query propagation is to use EIGRP stub routers where appropriate. When a router is single-homed to the rest of the network, and no other networks exist beyond that router, there is no real point in sending it a query looking for lost networks. The stub router declares itself as a stub to the router connected to the rest of the network, which is considered a hub router. The hub router then forwards no queries to the stub router because it knows there are no other networks, beyond those reported, existing beyond the stub router. In the topology for this lab, switch D1 is a stub router and R1 is its hub router.

- a. To verify that switch D1 is receiving EIGRP queries, issue the **shutdown** command on R2 interface E0/1. On switch D1, issue the **show ip eigrp events** command. This command outputs a timestamped list of actions that EIGRP is taking. In the output, you will find an entry that says switch D1 received a query trying to find the 10.23.0.0/24 network. Take note of the time stamp.

```
D1# show ip eigrp events
```

```
Event information for AS 98:
```

```
1 00:10:46.753 NDB delete: 10.23.0.0/24 1
2 00:10:46.753 RDB delete: 10.23.0.0/24 192.168.3.1
3 00:10:46.737 Metric set: 10.23.0.0/24 metric(Infinity)
4 00:10:46.737 Poison squashed: 10.23.0.0/24 lost if
5 00:10:46.737 Poison squashed: 10.23.0.0/24 rt net gone
6 00:10:46.737 Route installing: 10.23.0.0/24 192.168.3.1
7 00:10:46.737 Send reply: 10.23.0.0/24 192.168.3.1
8 00:10:46.737 Not active net/1=SH: 10.23.0.0/24 1
9 00:10:46.737 FC not sat Dmin/met: metric(Infinity) metric(1782272)
10 00:10:46.737 Find FS: 10.23.0.0/24 metric(1782272)
11 00:10:46.737 Rcv query met/succ met: metric(Infinity) metric(Infinity)
12 00:10:46.737 Rcv query dest/nh: 10.23.0.0/24 192.168.3.1
```

- b. Issue the **no shutdown** command on R2 interface E0/1.

- c. Configure D1 as an EIGRP stub router.

```
D1# config t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
D1(config)# router eigrp 98
```

```
D1(config-router)# eigrp stub
```

```
D1(config)# end
```

```
D1#
```

```
Mar 6 00:11:40.624: %DUAL-5-NBRCHANGE: EIGRP-IPv4 98: Neighbor 192.168.3.1
(Ethernet0/0) is down: peer info changed
```

```
Mar 6 00:11:45.174: %DUAL-5-NBRCHANGE: EIGRP-IPv4 98: Neighbor 192.168.3.1
(Ethernet0/0) is up: new adjacency
```

- d. Verify that R1 sees switch D1 as a stub by examining the output of the **show ip eigrp neighbor detail** command.

```
R1# show ip eigrp neighbor detail
```

```
EIGRP-IPv4 Neighbors for AS(98)
```

H	Address	Interface	Hold Uptime (sec)	SRTT (ms)	RTO	QSeq CntNum
2	192.168.3.2	E0/1	14 00:00:39	3	100	010

```
Version 25.0/2.0, Retrans: 1, Retries: 0, Prefixes: 3
Topology-ids from peer - 0
Topologies advertised to peer: base
```

```
Stub Peer Advertising (CONNECTED SUMMARY ) Routes
```

```
Suppressing queries
```

1	10.13.0.3	S1/0	11 00:11:44	1	100	028
---	-----------	------	-------------	---	-----	-----

```
Time since Restart 00:04:06
Version 25.0/2.0, Retrans: 0, Retries: 0, Prefixes: 3
Topology-ids from peer - 0
Topologies advertised to peer: base
```

```
0 10.12.0.2 Et0/0 12 00:12:24 1 100 030
Time since Restart 00:04:07
Version 25.0/2.0, Retrans: 2, Retries: 0, Prefixes: 2
Topology-ids from peer - 0
Topologies advertised to peer: base
```

```
Max Nbrs: 0, Current Nbrs: 0
```

- e. Issue the **shutdown** command on R2 interface E0/1. Take note of the timestamp on the syslog message reporting that the interface is down.
- f. On switch D1, issue the **show ip eigrp events** command. You will see that no query was received looking for the 10.23.0.0/24 network. The query was stopped at R1, speeding up the convergence process.

```
D1# show ip eigrp events
```

```
Event information for AS 98:
```

```
1 00:12:53.776 NDB delete: 10.23.0.0/24 1
2 00:12:53.776 Poison squashed: 10.23.0.0/24 rt net gone
3 00:12:53.776 RDB delete: 10.23.0.0/24 192.168.3.1
4 00:12:53.776 Not active net/1=SH: 10.23.0.0/24 1
5 00:12:53.776 FC not sat Dmin/met: metric(Infinity) metric(1782272)
6 00:12:53.776 Find FS: 10.23.0.0/24 metric(1782272)
7 00:12:53.776 Rcv update met/succmet: metric(Infinity) metric(Infinity)
8 00:12:53.776 Rcv update dest/nh: 10.23.0.0/24 192.168.3.1
9 00:12:53.776 Ignored route, hopcount: 10.23.0.0 255
10 00:12:50.077 Metric set: 172.16.3.0/24 metric(1910016)
11 00:12:50.077 Route installed: 172.16.3.0/24 192.168.3.1
12 00:12:50.077 Route installing: 172.16.3.0/24 192.168.3.1
13 00:12:50.077 Find FS: 172.16.3.0/24 metric(Infinity)
14 00:12:50.077 Free reply status: 172.16.3.0/24
15 00:12:50.077 Clr handle num/bits: 0 0x0
16 00:12:50.077 Clr handle dest/cnt: 172.16.3.0/24 0
17 00:12:50.077 Rcv reply met/succ met: metric(1910016) metric(1909760)
18 00:12:50.077 Rcv reply dest/nh: 172.16.3.0/24 192.168.3.1
19 00:12:50.077 Metric set: 10.23.0.0/24 metric(1782272)
20 00:12:50.076 Route installed: 10.23.0.0/24 192.168.3.1
21 00:12:50.076 Route installing: 10.23.0.0/24 192.168.3.1
22 00:12:50.076 Find FS: 10.23.0.0/24 metric(Infinity)
23 00:12:50.076 Free reply status: 10.23.0.0/24
```

- g. Issue the **no shutdown** command on R2 interface E0/1.

Step 4: Filter EIGRP routes with a distribute list.

EIGRP supports several different filtering capabilities. The simplest and most direct is to use a distribute list. A distribute list refers to an access list which can be applied to all EIGRP updates being sent by a certain router, or it can be applied to a specific interface to modify updates as they exit. For this exercise, we will filter the 10.12.0.0/24 network from updates being sent out of R2 interface E0/1. This will cause a change in R3's routing table.

- a. Examine the routing table on R3 by issuing the **show ip route eigrp | begin Gateway** command. In the output, you can see that R3 has calculated the path via R2 at 10.23.0.2 to be the best path to reach the 10.12.0.0/24 network.

```
R3# show ip route eigrp | begin Gateway
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
```

```
D    10.12.0.0/24 [90/307200] via 10.23.0.2, 00:00:17, Ethernet0/0
```

```
D    192.168.0.0/22 [90/3328] via 10.23.0.2, 00:00:53, Ethernet0/0
```

- b. Our intent is to change the configuration at R2 so that R3 only learns about the 10.12.0.0/24 network from R1. Create an access list that denies the 10.12.0.0/24 network and permits all other networks.

```
R2# config t
```

```
R2(config)# ip access-list standard EIGRP-FILTER
```

```
R2(config-std-nacl)# deny 10.12.0.0 0.0.255.255
```

```
R2(config-std-nacl)# permit any
```

```
R2(config-std-nacl)# exit
```

- c. Enter EIGRP router configuration mode and configure the distribute list to reference the access list you just created, further specifying that the filter should be effective outbound on interface E0/1.

```
R2(config)# router eigrp 98
```

```
R2(config-router)# distribute-list EIGRP-FILTER out e0/1
```

```
R2(config-router)# end
```

```
*Mar 6 00:19:56.379: %DUAL-5-NBRCHANGE: EIGRP-IPv4 98: Neighbor 10.23.0.3  
(Ethernet0/1) is resync: intf route configuration changed
```

- d. On R3, issue the **show ip route eigrp | begin Gateway** command. As you can see, the successor for the 10.12.0.0/24 network has changed to R1 at 10.13.0.1.

```
R3# show ip route eigrp | begin Gateway
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
```

```
D    10.12.0.0/24 [90/1792256] via 10.13.0.1, 00:01:30, Serial0/1/0
```

```
D    192.168.0.0/22 [90/3328] via 10.23.0.2, 00:03:00, Ethernet0/0
```